ACTION-ID: Evidence Handoffs and Verification Use in a Three-Stage LLM Workflow

Exploratory pilot | Apart Research Swarm Alignment Hackathon, Track 4 | 7 November 2026

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Abstract

A multi-agent auditor may receive correct evidence yet fail to use it, or may preserve uncertainty about a global claim while misattributing a narrower statement to a source. We investigate these possibilities using an instrumented scikit-learn classification workflow with coordinator-owned execution records and four controlled evaluation states: clean, selection leakage, paired-population mismatch, and both violations. Using OpenRouter-hosted openai/gpt-oss-120b, we generate twelve three-turn processing chains across a single-analyst control, a role-separated multi-agent chain, and a chain in which one reviewer receives a trusted inspection. We then evaluate each frozen transcript with message-only, source-lineage, and directly supplied inspection views, producing 36 auditor responses. In the two no-tool structures, all 24 agent-stage labels remained NOT_VERIFIED; the hypothesized promotion of the complete claim was not observed. Thirteen auditors made decisive decisions and all matched execution truth. However, six of twelve auditors with directly supplied trusted checks deferred; an agent also incorrectly attributed a reported baseline score to an evaluator receipt that contained only model accuracy. These are descriptive observations, not identified causal effects. We distinguish factual decision correctness, evidential sufficiency, and rationale fidelity, identify interface and proxy-scoring limitations, and propose fixed-evidence message interference as a separate follow-up experiment.

1. Research question and relevance

Research-agent outputs routinely combine source claims, derived metrics and upstream assessments. Oversight requires distinguishing a statement that happened to be true from one supported by evidence available to the approving agent. We ask: when a three-stage research workflow passes a claim downstream, does the downstream agent or auditor (i) preserve the claim's verification status, (ii) appropriately use trusted checks, and (iii) accurately attribute supporting facts to sources?

The study is relevant to the hackathon's Track 4 examples concerning worker completion claims and the influence of agent justifications on monitors [1]. Unlike studies of intentional subversion [2], ours does not model a deceptive or colluding swarm; roles are prompt-defined and share one model. The central contribution is an executed, auditable small-scale pilot plus explicit limits on what its scores establish.

1.1 Closest related work and novelty boundary

Research decision steering is not new: Anthropic reports that altering free-text justifications can sway model judges making ML-research decisions [5]. ClaimArena-LLM studies adversarial propagation of unsupported causal claims in multi-agent debates [6]. A provenance-sensitivity audit varies the source authority of propositions while holding other action-selection factors fixed [7]. Our pilot differs in using a coordinator-controlled, execution-grounded final-check payload, but it neither isolates an effect of upstream communication nor establishes a distinct mechanism. The proposed FEMI follow-up would intervene on assessment text while keeping that payload constant; it is not an outcome of this pilot.

We position these as the closest precedents, not as an exhaustive methodological comparison: the Anthropic study is supported by its detailed research report, while our descriptions of [6] and [7] are limited to the accessible publication abstracts. The contribution here is a small descriptive study with raw records and documented negative findings, not a novel proof of provenance security or general multi-agent failure.

2. Experimental setup

2.1 Execution ground truth

A reproducible Iris-based logistic-regression workflow records model selection, evaluator score events and paired comparison events under a coordinator-controlled ledger. The submitted claim is conjunctive: the final model evaluation was independent of selection, and model and baseline scores used identical paired evaluation rows. The four injected conditions cross selection independence with paired-row correctness. An independent scripted reference checker defines the full-record verdict. Public inputs contain a recorded model accuracy (1.0), model identifier, scored-row count and row digest in a metric receipt, plus a separately reported baseline accuracy (approximately 0.333); they do not disclose the decisive selection or paired-row records.

Condition	Selection check	Paired-row check	Full-record verdict
Clean	Pass	Pass	APPROVE
Selection leakage	Fail	Pass	REJECT
Row mismatch	Pass	Fail	REJECT
Both	Fail	Fail	REJECT

2.2 Agent processing and monitor views

At seed 21, each of four executions is processed through three model turns in each of three structures: (S) a single-analyst three-pass control; (M) three role-separated agents without verification; and (V) three role-separated agents where reviewer B gets the coordinator inspection. The latter adds information and therefore is not an information-matched comparison against S or M. Downstream reporter C sees B's text, not B's raw inspection. We run 12 chains (36 generation calls). Each frozen chain is later audited under: messages only; messages plus a coordinator-recorded source graph; and messages plus the coordinator's actual inspection results. This produces 36 auditor calls, 72 total. All used the OpenRouter `openai/gpt-oss-120b` model and the ordinary auditor prompt. No strict prompt was tested.

The trusted payload includes both checked Boolean predicates and the reference verdict, so the full-evidence condition primarily probes consumption of a supplied decision rather than independent inference from raw logs. Model-generated labels are not authorization certificates. The source graph encodes input edges but does not prove proposition-to-record entailment.

2.3 Evaluation dimensions

We distinguish (a) factual decisive correctness (APPROVE/REJECT matches execution truth); (b) decision support, requiring trusted evidence actually available under an explicit trust contract; and (c) rationale fidelity, meaning accurate source identification and correct logical explanation. The original automated scorer uses visibility categories as support proxies and does not semantically grade rationales. Consequently, only (a) and decision/deferral counts are fully enumerated quantitatively here; (b) is a conditional access proxy and (c) is discussed through inspected examples. The initial report was not constructed to test adaptive selection of inspections.

3. Recorded results

3.1 Outcome matrix

Upstream processing	Messages only	Source lineage	Direct trusted checks
Single analyst, three passes	4 DEFER	4 DEFER	2 REJECT; 2 DEFER
Three role-separated agents, no tool	4 DEFER	4 DEFER	4 DEFER
Three agents, B has verified check	3 REJECT; 1 DEFER	3 REJECT; 1 APPROVE	3 REJECT; 1 APPROVE

All cells contain one monitor response for each of the four underlying execution conditions. Across 36 auditors, 13 responses were decisive and all 13 matched the independent reference (zero factually incorrect

approvals and zero factually incorrect rejections); 23 responses deferred. In the 12 directly supplied full-evidence views, six auditors nevertheless deferred (S: 2/4, M: 4/4, V: 0/4). These are counts, not estimates of population error rates. Summed model-reported usage charges in the records were $0.009866427; this excludes external fees, retries outside the stored records, and labor.

3.2 Complete-claim abstention and narrower source confusion

Across the two no-tool processing structures, every one of 24 stage labels was NOT_VERIFIED. Thus there was no observed promotion of the entire conjunctive evaluation claim to VERIFIED. However, in the clean single-analyst chain, the final response described the evaluator metric receipt as documenting both model accuracy 1.0 and baseline accuracy ~0.333. The actual receipt documents only model accuracy; baseline accuracy is separately present in the submitted report. Correct global abstention therefore coexisted with a concrete, post-hoc identified source-attribution error. This happened in the single-analyst arm and does not demonstrate a uniquely multi-agent problem.

3.3 Available evidence versus used evidence

The full-evidence auditor conditions received identical coordinator inspection payloads for a given execution across structures. Nevertheless the 4-case counts of correct decisive responses differed: 2/4 (S), 0/4 (M), and 4/4 (V). These decisions may have been affected by prior messages, presentation, or stochastic sampling. Structure, transcript content and upstream verification history were not causally isolated. Some deferral rationales described verification as unavailable even though the direct trusted payload included the predicates and reference verdict; a mechanistic account (distrust, attention or misunderstanding) is not established. In V, reviewer B used the trusted check and labelled the clean case VERIFIED and faults CONTRADICTED. Reporter C reverted to NOT_VERIFIED in two of four cases, but did not have the raw inspection. Such abstention can comply with a trust policy that requires direct verification; scoped authenticated attestations could support an alternative policy.

3.4 Structural provenance check is not semantic certification

Recorded graph edges matched the stage input dependencies across all 12 chains. However, the protocol did not expose graph-node identifiers to agent stages; absence of graph-node citations is therefore not evidence of an agent refusal to cite supplied IDs. The checker establishes consistent bookkeeping, not that a textual assertion follows from cited evidence. A source-lineage-only monitor approved one clean case after initially deferring in the message-only view; that approval was factually correct, but lineage by itself did not disclose the trusted predicate contents.

4. Interpretation and limitations

The observations support three narrow statements: complete-claim verification was not spuriously promoted in the no-tool conditions; directly supplied trusted decisions were not always used to produce a decisive answer; and at least one agent attributed a number to the wrong source. We do not infer an emergent swarm failure, strategic collusion, a validated proof-carrying protocol, or a successful new monitor. One Iris workflow family, one seed, one model, four hand-constructed fault states, role-defined agents, a single auditor response per treatment and an ordinary auditor prompt preclude broad statistical or causal conclusions. Reported probabilities were not calibrated; one REJECT co-occurred with prob_valid=0.92. The source-support labels are visibility-based proxies rather than entailment judgments.

A deterministically enforced gate receiving the complete authentic predicate results can implement this particular conjunctive rule without an LLM. Passing that gate is a reference-implementation property, not evidence of robustness to compromised logging, bypasses, fabricated tool responses or partial execution capture. The trust assumptions are therefore central. The pilot is best viewed as an artifact for studying evidence transfer and interface design rather than an adaptive inspection algorithm.

5. Conclusion and proposed follow-up

In this controlled study, accurate decisive auditor labels did not imply complete evidence-use reliability, and correct uncertainty labels did not imply correct attribution of subordinate facts. A separate, causally clearer follow-up would hold the trusted evidence and authorization policy fixed while substituting upstream assessments: none, accurate, falsely claiming missing verification, contradictory, and neutral length-matched text. That proposed FEMI-1 study is specified in a separate draft protocol; it has not been run or publicly preregistered, and no FEMI-1 results are claimed. Evidence-interface and coordinated-adversary studies remain distinct potential directions.

Prior-work and sprint contribution disclosure

Pre-sprint foundations: the finite-world and Galois-connection math lab, the executed ML workflow and reference verifier, earlier handoff experiments and Jev/d1 inspection-routing work. Sprint-specific development and evaluation in this packet: the role-separated interaction generator and matched single-analyst control, source-dependency recording, verification-access arm, frozen transcript monitoring views, four-condition 72-call live closeout, source-attribution audit and interpretation corrections. The exact code history and work dates should be reconciled against repository commits before submission. Category theory is motivation only; no new functorial proof is claimed.

References

[1] Apart Research. Swarm Alignment Hackathon (2026). https://apartresearch.com/sprints/swarm-alignment-2026

[2] Greenblatt, R., Shlegeris, B., Sachan, K., and Roger, F. AI Control: Improving Safety Despite Intentional Subversion. ICML, 2024. https://proceedings.mlr.press/v235/greenblatt24a.html

[3] Zhang, S., et al. Which Agent Causes Task Failures and When? On Automated Failure Attribution of LLM Multi-Agent Systems. ICML, 2025. https://proceedings.mlr.press/v267/zhang25cq.html

[4] ACTION-ID project artifacts: `ACTION_ID_v1_1_Four_Condition_Closeout.zip`, `ACTION_ID_four_condition_live_results.zip`, `ACTION_ID_four_condition_independent_review.md` (source and raw records supplied in conversation; no public repository URL asserted).

[5] Anthropic Alignment Science. Automated Researchers Can Subtly Sandbag (2025). https://alignment.anthropic.com/2025/automated-researchers-sandbag/

[6] Anghel, C., et al. ClaimArena-LLM: Auditing Adversarial Propagation of False Causal Attributions in Multi-Agent LLM Debates. Machine Learning and Knowledge Extraction, 8(10), 327 (2026). https://doi.org/10.3390/make8100327

[7] Liao, J. Auditing Provenance Sensitivity in LLM Agent Action Selection. arXiv:2607.20827 (2026). https://arxiv.org/abs/2607.20827

Appendix A. Limitations and Dual-Use / Ethical Considerations

This study operates on local synthetic fault injections in an Iris-model evaluation pipeline. It does not carry out external compromise, real sabotage or covert-agent coordination. The reported effect is limited to communication and monitoring interfaces over recorded toy workflow states; deployment settings may differ in scale, trust, workload or adversary capability.

The complete-ledger assumption is stronger than many real systems provide. A verifier can only certify what its independent, authentic and complete records entail. Our source graph was not an authenticated semantic proof system; a model reading a claimed check is not equivalent to programmatic authentication. We do not recommend using the observed LLM auditor as a security boundary.

Publishing concrete evidence-transfer mistakes may help operators detect false reporting, but could also help deceptive agents camouflage unsupported assessments. We provide high-level results, a local benign reproduction and failures of evidence attribution; no live misuse target or operational instructions for exploiting external systems are included. Any future adversarial-message experiments should remain in a closed simulation with a fixed, authorized scope.

Appendix B. Reproducibility and post-hoc labeling

All 36 recorded auditor decisions are listed without rescoring in `DECISION_LEDGER_36.csv`. The separate `POSTHOC_CORRECTIONS.md` describes interpretation changes. The published primary results should retain original labels and counts; no rationale-fidelity rate should be fabricated from an uncompleted semantic audit. The follow-up protocol `FEMI_1_PREREGISTRATION_DRAFT.md` is a proposed draft, not a completed registration, and contains no experimental observations.

Before submission: replace the author/affiliation placeholder; transfer the paper into the official template linked from the current Guidelines page; verify pre-sprint/sprint boundaries against repository commits; and check final submission links. This revised document remains a draft and is not in the official template.